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IT Project Management Support

Lecture 7

Earned Value Management

Literature

- Project Management Institute. A Guide to the Project Management Body of Knowledge (PMBOK Guide) – 6th Edition 2017
- Kathy Schwalbe: Information Technology Project Management, 8th Edition. Cengage Learning 2016
- Earned Value Primer: The Basics of EVM by Dr Mike Clayton:
<https://onlinepmcourses.com/earned-value-primer/>
- Earned Value Tutorial:
<https://www.projectengineer.net/tutorials/earned-value-tutorial/> accessed on August 10, 2020
- What is Earned Value? by Duncan Haughey:
<https://www.projectsmart.co.uk/what-is-earned-value.php> accessed on August 10, 2020

Earned Value Management (EVM)

- Earned Value Management (EVM) is a project management technique for measuring project performance and progress in an objective manner.
- Earned value management is a project management technique for measuring project performance and progress.
- It has the ability to combine measurements of the project management triangle: scope, time, and costs.
- In a single integrated system, earned value management is able to provide accurate forecasts of project performance problems, which is an important contribution for project management.

Earned Value Analysis (EVA)

- **Earned Value Analysis (EVA)** is a method that allows the project manager to measure the amount of work actually performed on a project beyond the basic review of cost and schedule reports.
- EVA provides a method that permits the project to be measured by progress achieved.
- The project manager is then able, using the progress measured, to forecast a project's total cost and date of completion, based on trend analysis or application of the project's "burn rate".
- This method relies on a key measure known as the project's **Earned Value**.

What is Earned Value?

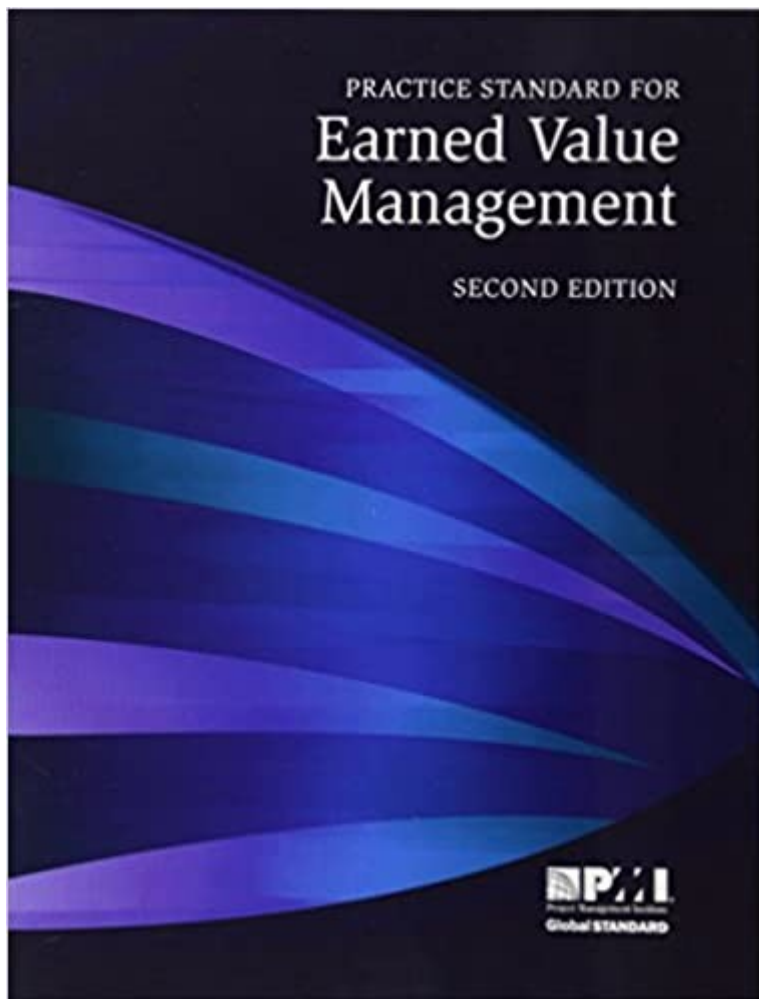
- Current performance is the best indicator of future performance, and, therefore, using trend data, it is possible to forecast cost or schedule overruns at an early stage in a project.
- The most comprehensive trend analysis technique is the **Earned Value Method**.
- In a nutshell, **Earned Value** is an approach where you monitor the project plan, actual work, and work completed value to see if a project is on track.
- **Earned Value** shows how much of the budget and time should have been spent, considering the amount of work done so far.

What is Earned Value?

- Here are five other definitions:
 1. Englert and Associates, Inc. define it as, *"a method for measuring project performance. It compares the amount of work that was planned with what was actually accomplished to determine if cost and schedule performance is as planned."*
 2. Project Magazine defines it as, *"a methodology used to measure and communicate the real physical progress of a project taking into account the work complete, the time taken and the costs incurred to complete that work."*
 3. The user guide for Microsoft Project 2003 defines Earned Value as, *"a method for measuring project performance. It indicates how much of the budget should have been spent, in view of the amount of work done so far and the baseline cost for the task, assignment, or resources."*

What is Earned Value?

4. Field Operative defines it as, *"the physical work accomplished plus the authorised budget for this work. The sum of the approved cost estimates, (which may include overhead allocation), for activities, (or portions of activities), completed during a given period, usually project-to-date."*
5. NASA defines it as, *"an integrated management control system for assessing, understanding and quantifying what a contractor or field activity is achieving with program dollars. EVM provides project management with objective, accurate and timely data for effective decision making."*



Practice Standard for Earned Value Management – Second Edition (2011)

Earned Value Management (EVM), known as “management with the lights on”, is based on the principle that past patterns and trends can indicate future conditions. EVM helps you clearly and objectively see where your project is headed compared to where it’s supposed to be.

The EVM methodology addresses project scope, schedule, and costs, and applies to many of the PMBOK® Guide Knowledge Areas and Process Groups.

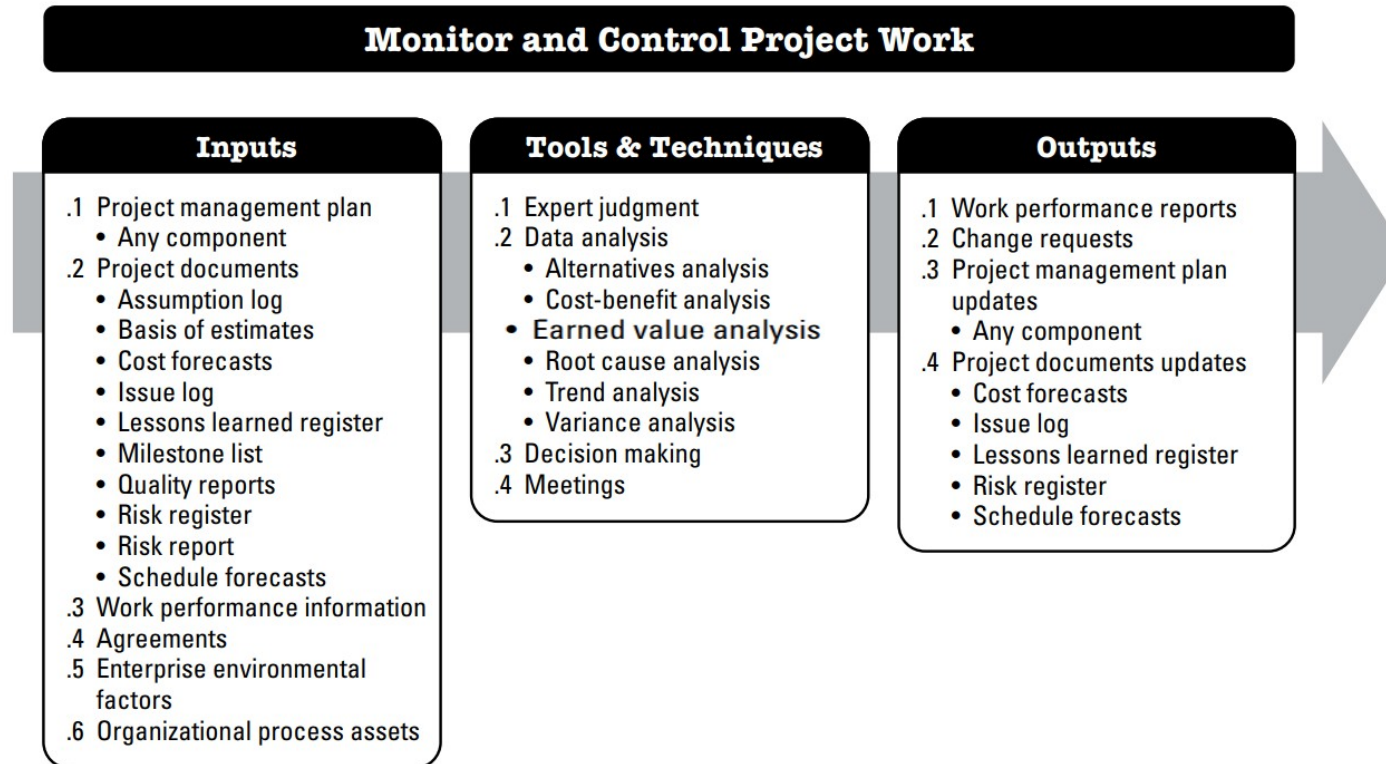
The Practice Standard for Earned Value Management – Second Edition is intended for any practitioner who wants to expand their toolset and use EVM to improve project performance.

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Monitor and Control Project Work

- Monitor and Control Project Work is the process of tracking, reviewing, and reporting the overall progress to meet the performance objectives defined in the project management plan.
- The key benefits of this process are that it allows stakeholders to understand the current state of the project, to recognize the actions taken to address any performance issues, and to have visibility into the future project status with cost and schedule forecasts.

The inputs, tools and techniques, and outputs of the process are depicted in Figure below.



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Control Schedule

Inputs

- .1 Project management plan
 - Schedule management plan
 - Schedule baseline
 - Scope baseline
 - Performance measurement baseline
- .2 Project documents
 - Lessons learned register
 - Project calendars
 - Project schedule
 - Resource calendars
 - Schedule data
- .3 Work performance data
- .4 Organizational process assets

Tools & Techniques

- .1 Data analysis
 - Earned value analysis
 - Iteration burndown chart
 - Performance reviews
 - Trend analysis
 - Variance analysis
 - What-if scenario analysis
- .2 Critical path method
- .3 Project management information system
- .4 Resource optimization
- .6 Leads and lags
- .7 Schedule compression

Outputs

- .1 Work performance information
- .2 Schedule forecasts
- .3 Change requests
- .4 Project management plan updates
 - Schedule management plan
 - Schedule baseline
 - Cost baseline
 - Performance measurement baseline
- .5 Project documents updates
 - Assumption log
 - Basis of estimates
 - Lessons learned register
 - Project schedule
 - Resource calendars
 - Risk register
 - Schedule data

Control Costs

Inputs

- .1 Project management plan
 - Cost management plan
 - Cost baseline
 - Performance measurement baseline
- .2 Project documents
 - Lessons learned register
- .3 Project funding requirements
- .4 Work performance data
- .5 Organizational process assets

Tools & Techniques

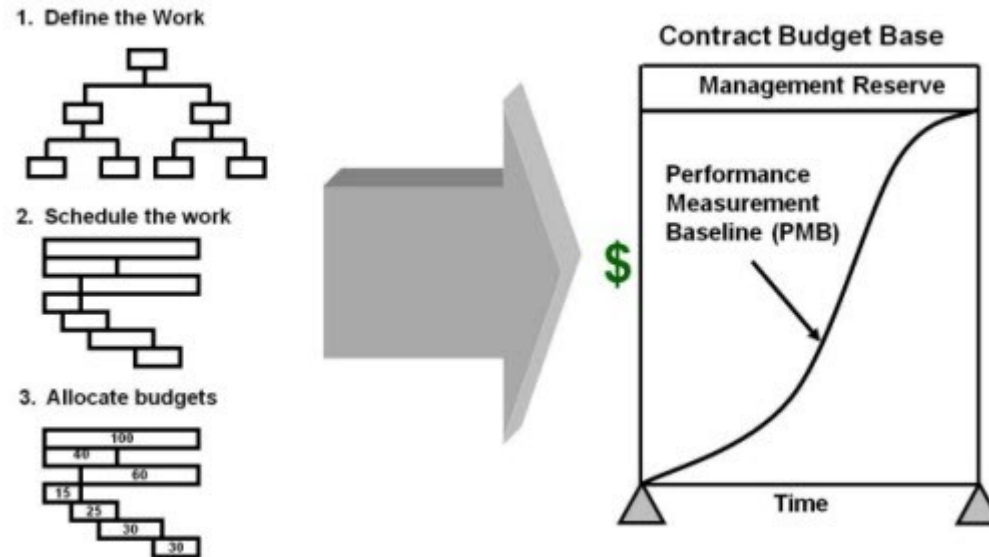
- .1 Expert judgment
- .2 Data analysis
 - Earned value analysis
 - Variance analysis
 - Trend analysis
 - Reserve analysis
- .3 To-complete performance index
- .4 Project management information system

Outputs

- .1 Work performance information
- .2 Cost forecasts
- .3 Change requests
- .4 Project management plan updates
 - Cost management plan
 - Cost baseline
 - Performance measurement baseline
- .5 Project documents updates
 - Assumption log
 - Basis of estimates
 - Cost estimates
 - Lessons learned register
 - Risk register

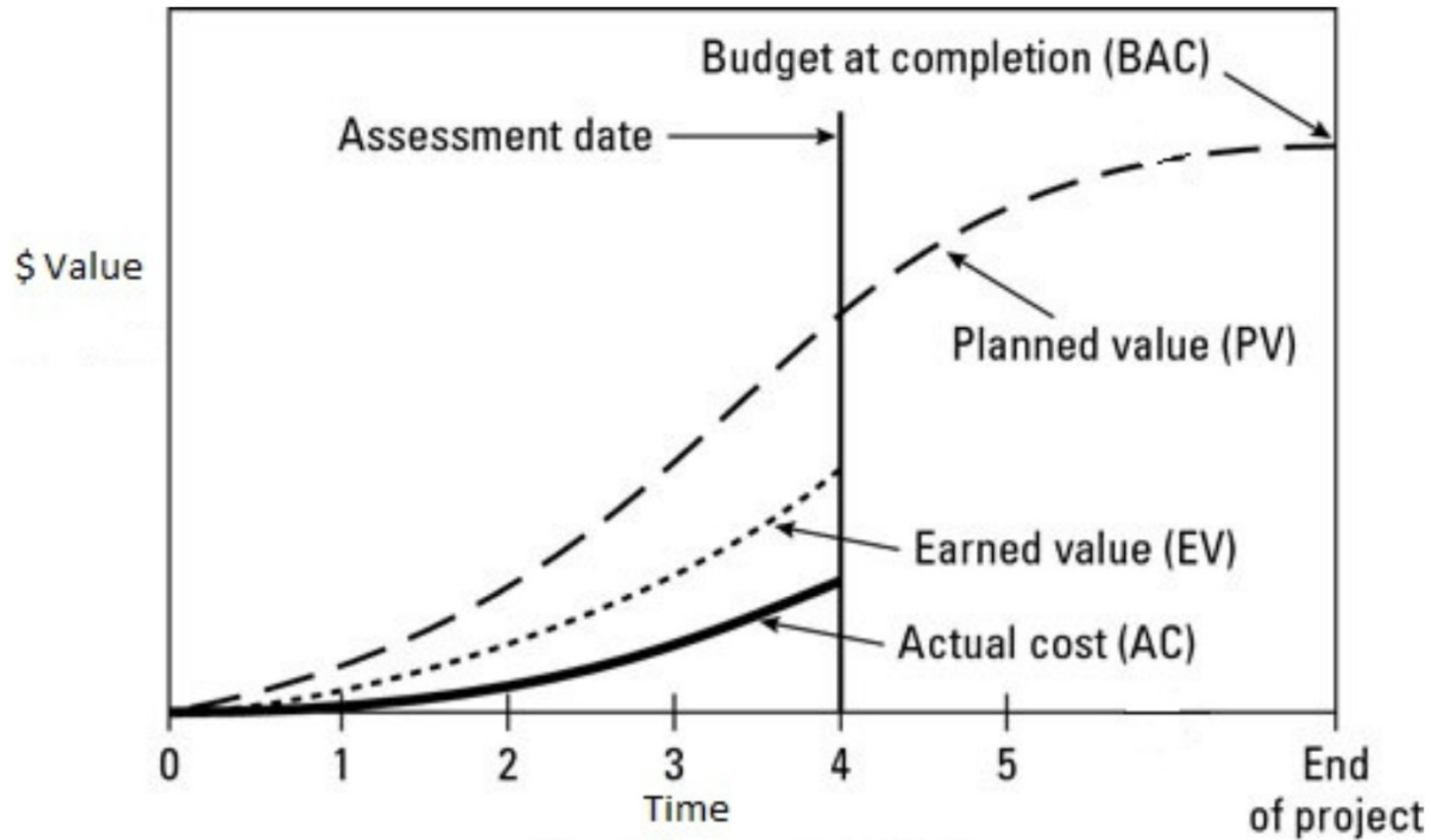
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Planning, Scheduling, and Budgeting



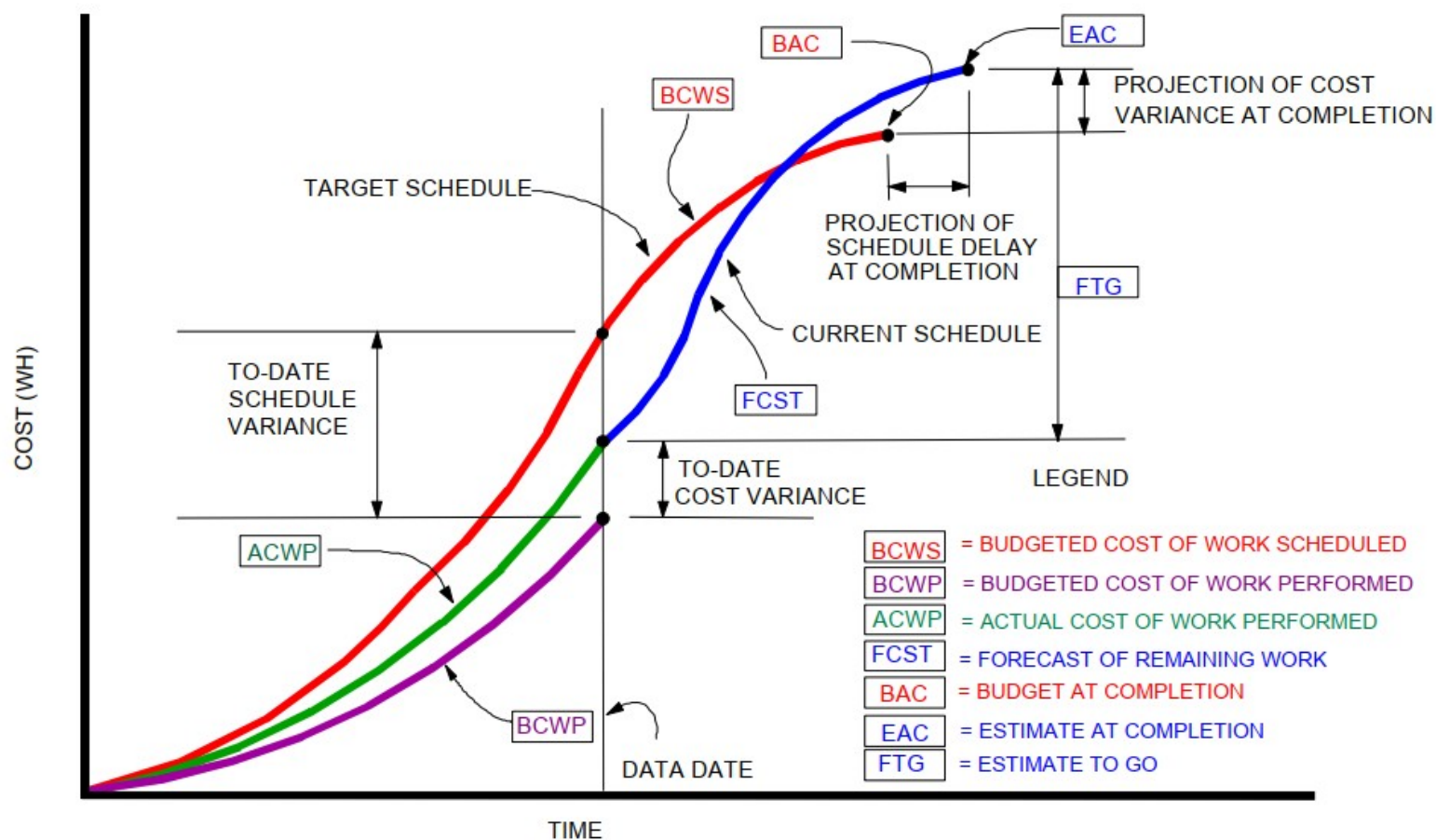
- The integrated master schedule is the project's road map to meet contract objectives.
- This schedule must be resource loaded to determine the budget for the work as scheduled.
- The resource loaded schedule is the basis for the monthly budget, or BCWS, for each task and thus the project.
- This time phased budget is the performance measurement baseline (PMB).
- The total budget for each task, control account, or the entire project is defined as the budget at complete (BAC).
- Because most projects are initiated with some level of uncertainty; i.e. risk, project managers typically set aside a portion of the total project value as a management reserve (MR). .

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PERFORMANCE MEASUREMENT DATA ELEMENTS



Basic Questions

- What's the best way to monitor and report on the performance of your projects?
- And how can you predict their future performance?
- The answer is Earned Value Analysis (EVA),
- and the process of working with Earned Value as your performance management tool is called Earned Value Management (EVM).

Why do We Need Earned Value?

- The monitoring and control cycle is of key importance for the project execution phase. So, for the execution to work well, we need to be able to:
 - Measure status
 - Assess issues
 - Design interventions
 - Communicate status for accountability and decision-making.

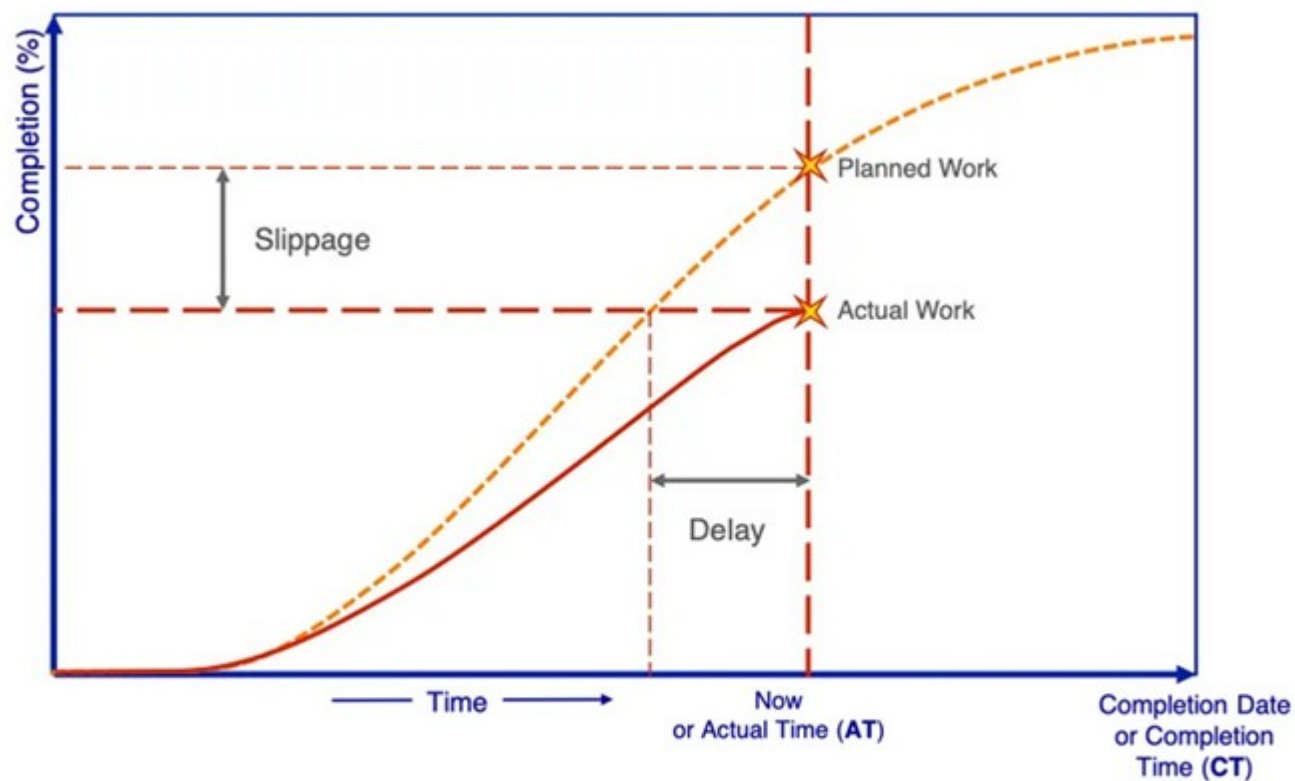
Why do We Need Earned Value?

- Earned Value Management provides us with a way to do this. It lets us:
 - Measure progress against plan and hence understand the status of your project
 - Assess and communicate performance with a set of standard performance measures
 - Estimate time and cost to complete work, against your plan
- You can only manage and control what you can measure, so measurement tools are always at a premium.
- And reporting progress may have a value, but better still is your ability to predict the future from trends.
- These are the tools that EVM Offers you.

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Schedule or Cost Performance?

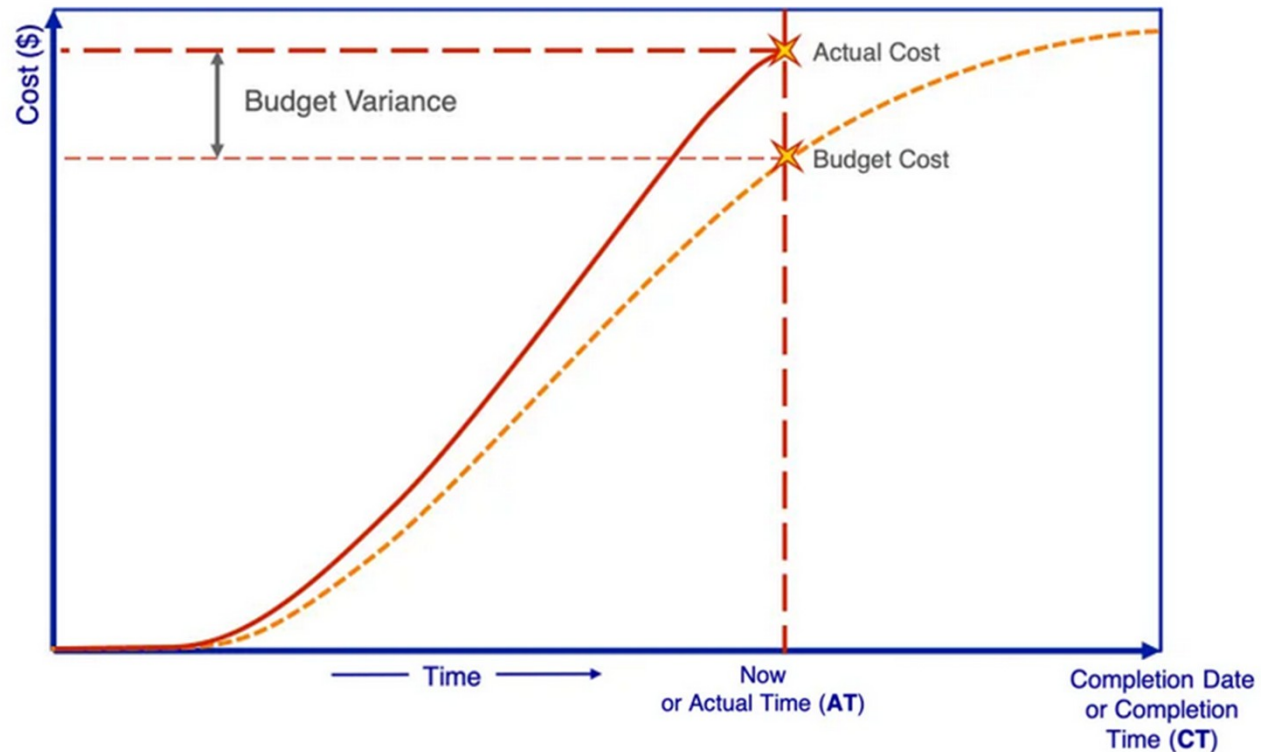
It is pretty straightforward to measure schedule performance...



Schedule Performance Chart

Schedule or Cost Performance?

And it is equally straightforward to measure budget performance...



Budget Performance Chart

But, what if you are late and under budget?

Is that good, or bad?

- If you could express budget and schedule performance in one language, you could combine the two.
- And that is exactly what Earned Value Analysis does.
- It uses an ‘S’ curve for the percentage complete.
- It looks at how well you have spent your budget on efficiently producing value.
- The value of the work you have completed to date, based on your original budget, is the Earned Value.

What is Earned Value Management?

- Earned Value Management (or EVM) is one of the most important techniques for formal project management.
- It will equip you to monitor and forecast robustly.
- So, if you aspire to lead a substantial project, you'll need to know when and how to deploy this powerful toolset.

What is Earned Value?

- Earned Value is the value of the work you have actually completed at the present time, based on your original budget.
- So, an alternative name for the Earned Value (EV) is the 'Budgeted Cost of Work Performed' or BCWP.

What is Earned Value?

“ *Earned Value is the value of the work you have actually completed at the present time, based on your original budget.*

So, an alternative name for the Earned Value (EV) is the ‘**Budgeted Cost of Work Performed**’ or BCWP.

Compare it with the **Actual Cost of Work Performed**. This is usually just referred to as the Actual Cost (AC).

A simple measure of the Earned Value assumes that we build out value in proportion to the amount of work completed. This gives us the formula:

“ *Earned Value = BAC x Percentage Completed*
BAC = Budget at Completion: the budgeted cost of the completed project.

The Key Earned Value Management Measures

- With BAC, Earned Value and Actual Cost, we have three of the four key EVM measures. The fourth is the **Planned Value (PV)**.
- **Planned Value** is the budget cost of the work you planned to have finished at the current time. So, it is also known as the **Budgeted Cost of Work Scheduled (BCWS)**.

So, there are four key parameters in EVM, that you can measure:

- **Planned Value (PV)** is the Budgeted Cost of Work Scheduled (BCWS)
- **Earned Value (EV)** is the Budgeted Cost of Work Performed (BCWP)
- **Actual Cost (AC)** is the Actual Cost of Work Performed (ACWP)
- Budgeted Cost of your project is the **Budget at Completion (BAC)**.

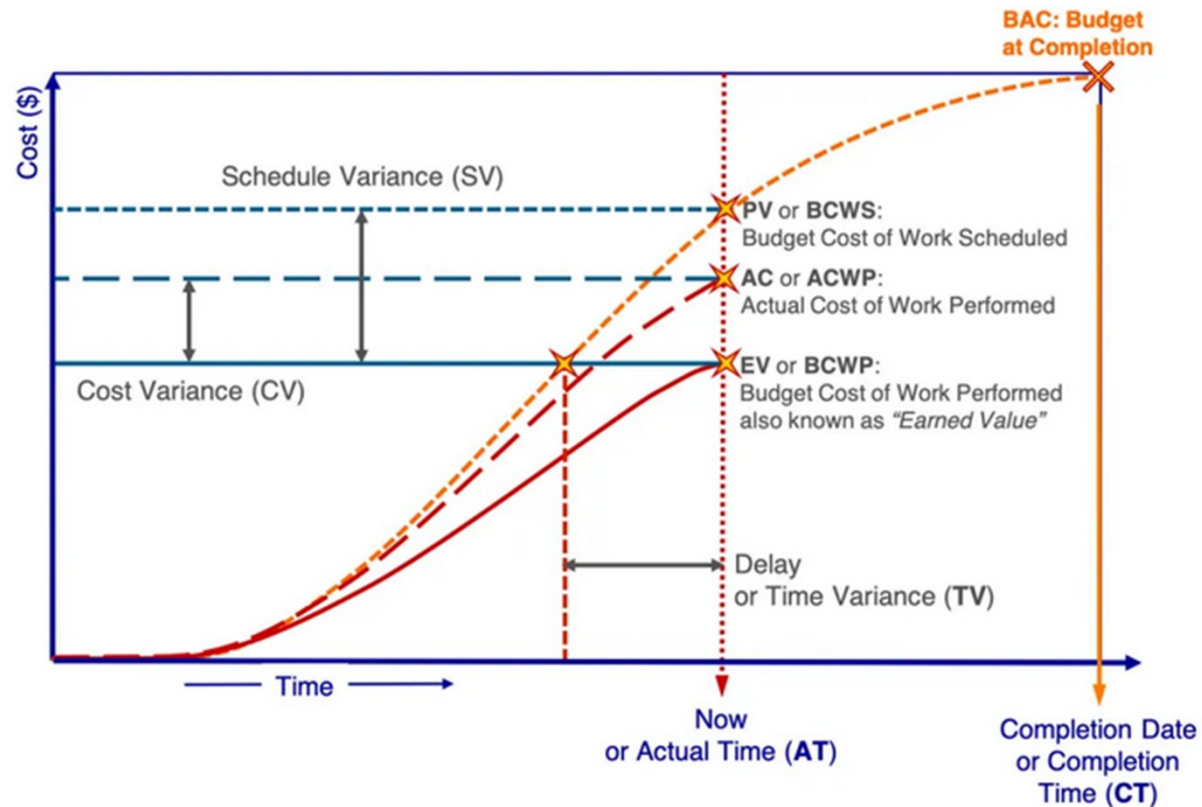
Measuring Performance with Earned Value

There are four standard performance measures in EVM:

- Cost Variance: **$CV = EV - AC$**
- Cost Performance Index: **$CPI = EV/AC$**
- Schedule Variance: **$SV = EV - PV$**
- Schedule Performance Index: **$SPI = EV/PV$** .

Variances from Plan

You can see what the Schedule Variance and Cost Variance look like on a chart...



Earned Value Analysis Chart

*Note that the Schedule Variance is expressed in units of money – it might perhaps be better labeled ‘**schedule-related cost variance**’.*
*In turn, the true time slippage is shown as the **Time Variance (TV)**.*

Performance Indices

- The performance indices express the variances as ratios.
- These allow them to be compared from one project to another.
- But it's useful to take a couple of examples, to better understand them.

Cost Performance

- A CPI of less than one, or a negative CV, indicates your project is under-performing against you plan on cost. That's a bad thing.
- A CPI of greater than one, or a positive CV, indicates your project is performing better than you planned on cost.
- A CPI value of 0.85 means that, for every dollar you have spent on the project to date, the project has only delivered US\$0.85 in Earned Value.

Schedule Performance

- A SPI greater than one, or a positive SV, indicates you have accomplished more work than you had planned.
- But this is important: an SPI of more than one does not necessarily mean you are ahead of schedule!
- You can accomplish more work than planned by working on non-critical path Work Packages.
- You need to look at the critical path to determine whether you are ahead, on or behind schedule.
- A SPI value of 1.15 means that for every dollar of work the you had planned to deliver at the current point in time, you have actually delivered US\$1.15 worth of work.

RAG or Traffic Light Reporting

- At the start of your project, it is wise to set levels of either performance index or variation, at which alarm bells will ring. A good approach is a traffic light system where:
- **Green** represents everything is within a narrow margin of your plan.
- **Amber** represents a significant level of concern that merits review and active management.
- **Red** represents a serious concern that requires reporting upwards and a substantial intervention.

RAG or Traffic Light Reporting

- For example...

	CPI	CV	SPI	SV
Green	>0.95	<5% of BAC	>0.95	<5% of CT
Amber	0.80 – 0.95	5% – 10% of BAC	0.80 – 0.95	5% – 10% of CT
Red	<0.80	>10% of BAC	<0.80	>10% of CT

With a RAG status for each performance index, you can highlight quickly the need for action, and also report efficiently to your steering group.

Forecasting with Earned Value Analysis

Earned Value Analysis forecasts are designed to make cost estimates, rather than time estimates.

The principal measure is the **Estimate at Completion (EAC)**. This compares directly to the **Budget at Completion (BAC)**.

Therefore:

Forecast **Cost Variance at Completion** **$VAC = BAC - EAC$**

A positive result (>0) suggests you anticipate completing under budget (very good).

A result less than zero anticipates a cost over-run.

So, crucially, how do we calculate our Estimate at Completion?

Estimate at Completion (EAC)

There are different ways to calculate your EAC.

Which you choose will depend upon the assumption you choose to make, about how your project will proceed from now on.

There are five common approaches:

1. Optimistic Approach: Reversion to Plan
2. Simple Approach: Continue as Before
3. Hybrid Approach
4. Pessimistic Approach: Worst of Both Worlds
5. Super-Hybrid Approach: Most Sophisticated

Estimate at Completion (EAC)

1. Optimistic Approach: Reversion to Plan

This approach assumes that from now on you will continue according to your original plan.

How likely is that? If you already have a substantial underperformance, this could represent a serious or willful decision to ignore that fact.

So, before you use this approach, ask: ‘what evidence do we have that things will change?’

The formula for this is:

$$\text{EAC} = \text{AC} + (\text{BAC} - \text{EV})$$

We start from the actual cost, and then add on the remaining budgetted cost, having delivered the Earned Value.

Estimate at Completion (EAC)

2. Simple Approach: Continue as Before

This approach assumes that you will continue to deliver the same level of cost performance as you have to date.

It is easy to calculate:

$$\text{EAC} = \text{BAC} / \text{CPI}$$

We divide the original budget cost by the cost performance index to date.

Estimate at Completion (EAC)

3. Hybrid Approach

$$\text{EAC} = \text{AC} + [(\text{BAC} - \text{EV}) / \text{CPI}]$$

We divide the outstanding work by the CPI and add the Actual Cost of the work performed.

Estimate at Completion (EAC)

4. Pessimistic Approach: Worst of Both Worlds

This approach factors in both the CPI and the SPI. It typically produces a more pessimistic (conservative) estimate.

$$\text{EAC} = \text{BAC} / (\text{CPI} \times \text{SPI})$$

This is like the ‘simple approach’ formula, but factors the BAC down twice – by each of the two indices.

Estimate at Completion (EAC)

5. Super-Hybrid Approach: Most Sophisticated

But, sophistication does not always give a better answer.

It is a variant on the pessimistic approach above, but gives a slightly different answer.

Beware the precision trap: precise is not the same as accurate.

$$\text{EAC} = \text{AC} + [(\text{BAC} - \text{EV}) / (\text{CPI} \times \text{SPI})]$$

This formula banks the cost performance to date (AC) and applies the factoring only to the remaining work.

Which one is correct?

We can quote the PMBoK on this point...

Each of these approaches can be correct for any given project and will provide the project management team with an ‘early warning’ signal if the EAC forecasts are not within acceptable tolerances.

The PMI Project Management Body of Knowledge (PMBoK) 6th edition

The PMBoK describes approaches 1, 2, and 5.

Estimate to Complete

Estimate to Complete (ETC) is a measure of the cost you now expect to spend, in completing the project.

It is your EAC less the amount you have already spent:

$$\text{ETC} = \text{EAC} - \text{AC}$$

To-Complete Performance Index

The To-Complete Performance Index (TCPI) is not a forecast, but a performance index, but we needed to understand the forecasts, before looking at this.

Perhaps unsurprisingly, there are two ways of calculating this, depending on whether you want to base it on your BAC or (to my preference) your EAC.

$$\text{TCPI (BAC)} = (\text{BAC} - \text{EV}) / (\text{BAC} - \text{AC})$$

$$\text{TCPI (EAC)} = (\text{BAC} - \text{EV}) / (\text{EAC} - \text{AC})$$

This is the ratio of the value remaining (BAC-EV) to the cost remaining based on either your budget (BAC-AC) or your estimate at completion (EAC-AC).

Carrying Out Earned Value Management

Setting-up Your Earned Value Management

Fundamentally, setting up EVM is a five-step process:

1. Establish your Project Scope.
2. Create your Work Breakdown Structure (WBS).
3. Identify dependencies and estimate durations to create your project schedule.
4. Allocate a budget to each activity or Work Package in your WBS. Do this by estimating the costs of people, materials, assets, and overheads.
5. Build your EVA spreadsheet, combining the budget and schedule data with spaces for monitoring and cells that calculate the indices, variances, and forecasts you will need. *It is recommended to use conditional formatting to highlight the status of RAG forecasts and indices.*

Carrying Out Earned Value Management

Measuring Progress

The challenge with measurement is to build a process that is consistent with the scale of expenditure and the risks inherent in your project.

You will also want to consider:

- size of work packages
- procurement contract terms
- payment timing
- milestones and gateways

You are looking to balance ease of measurement with accuracy, and to find a solution that all your key stakeholders can accept as the basis for future project decision-making and evaluation.

You can base your measurement on percentage completion approach.

Examples of Percentage Completion

	<i>Actual Percentage Complete</i>	<i>Reported Percentage Complete</i>
Two-Point	<100%	0%
	100%	100%
Three Point	<33%	0%
	33-67%	33%
	>67%	67%
	100%	100%
Four-Point	0-25%	0%
	25-50%	25%
	50-75%	50%
	75-100%	75%
	100%	100%

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Summary of Earned Value Analysis Formulas

Primary Earned Value Parameters

Budgeted Cost of your project is the Budget at Completion (BAC)

Planned Value (PV) is the Budgeted Cost of Work Scheduled (BCWS)

Earned Value (EV) is the Budgeted Cost of Work Performed (BCWP)

$$\text{Earned Value} = \text{BAC} \times \text{Percentage Completed}$$

Actual Cost (AC) is the Actual Cost of Work Performed (ACWP)

Earned Schedule (ES) is the Schedule Time of Work Performed (STWP)

Summary of Earned Value Analysis Formulas

Earned Value Performance Measures

Cost Variance: $CV = EV - AC$

Cost Performance Index: $CPI = EV/AC$

Schedule Variance: $SV = EV - PV$

Schedule Performance Index: $SPI = EV/PV$

Summary of Earned Value Analysis Formulas

Earned Value Forecasts

Forecast Cost Variance at Completion $VAC = BAC - EAC$

Estimate at Completion Formulas

$$EAC = AC + (BAC - EV) \quad *$$

$$EAC = BAC / CPI \quad *$$

$$EAC = AC + [(BAC - EV) / CPI]$$

$$EAC = BAC / (CPI \times SPI)$$

$$EAC = AC + [(BAC - EV) / (CPI \times SPI)] \quad *$$

* Included in the PMBoK 6th Edition. You need these for CAPM and PMP exams.

Estimate to Complete (ETC) = $EAC - AC$

To-Complete Performance Index

$$TCPI (BAC) = (BAC - EV) / (BAC - AC)$$

$$TCPI (EAC) = (BAC - EV) / (EAC - AC)$$



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Thank You very much
for your attention